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A Reevaluation of Impact Melt Production

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The production of melt and vapor is an important process in impact cratering events. Because significant melting and vaporization do not occur in impacts at velocities currently achievable in the laboratory, a detailed study of the production of melt and vapor in planetary impact events is carried out with hydrocode simulations.

Sandia's two-dimensional axisymmetric hydrocode CSQ was used to estimate the amount of melt and vapor produced for widely varying initial conditions: 10 to 80 km/sec for impact velocity, 0.2 to 10 km for the projectile radius. Runs with different materials demonstrate the material dependency of the final result. These results should apply to any size projectile (for given impact velocity and material), since the results can be dynamically scaled so long as gravity is unimportant in affecting the early-time flow.

In contrast with the assumptions of previous analytical models, a clear difference in shape, impact-size dependence, and depth of burial has been found between the melt regions and the isobaric core. In particular, the depth of the isobaric core is not a good representation of the depth of the melt regions, which form deeper in the target. While near-surface effects cause the computed melt region shapes to look like "squashed spheres" the spherical shape is still a good analytical analog.

One of the goals of melt production studies is to find proper scaling laws to infer melt production for any impact event of interest. We tested the point source limit scaling law for melt volumes (μ = 0.55–0.6) proposed by M. D. Bjorkman and K. A. Holsapple (1987, *Int. J. Impact Eng.* 5, 155–163). Our results indicate that the point source limit concept does not apply to melt and vapor production. Rather, melt and vapor production follows an energy scaling law (μ = 0.67), in good agreement with previous results of T. J. Ahrens and J. D. O'Keefe [1977, in *Impact and Explosion Cratering* (D. J. Roddy, R. O. Pepin, and R. B. Merrill, Eds.), pp. 639–656, Pergamon Press, Elmsford, NY].

Finally we tested the accuracy of our melt production calculation against a terrestrial dataset compiled by R. A. F. Grieve and M. J. Cintala (1992, *Meteorities* 27, 526–538). The hydrocode melt volumes are in good agreement with the estimated volumes of that set of terrestrial craters on crystalline basements. At present there is no good model for melt production from impact craters on sedimentary targets. © 1997 Academic Press

INTRODUCTION

Melt and vapor production in impact cratering events affects the understanding of many planetary geology problems, from the formation of complex craters (Grieve and Cintala 1992), to the effects of catastrophic impact events (Tonks and Melosh 1993), to the climatic and environmental consequences of impact cratering (Toon *et al.* 1994). Even though there have been many studies on the subject (e.g., O'Keefe and Ahrens 1977, Orphal *et al.* 1980, Croft 1982, Bjorkman and Holsapple 1987, Grieve and Cintala 1992, Tonks and Melosh 1993) we are still far from a complete understanding of the problem. Experimental approaches, while highly desirable, are limited because the maximum impact velocities currently obtained in the laboratory (6–8 km/sec) do not reach the magnitude of planetary impact events (≥20 km/sec). Consequently only theoretical studies can address the problem at the present time. Two general approaches are possible: (1) analytical energy balance models, and (2) numerical solutions of the fundamental dynamic equations. Both approaches rely on the differential equations established through the conservation of mass, momentum, and energy and on the equation of state of the materials making up the projectile and target. While analytical models require drastic simplification in the equations and their integration, numerical simulations use computer codes to numerically integrate the original differential equations, and need sophisticated computer algorithms and long run times to be successful. The conflicting needs for simplicity and ease of calculation versus the need for accurate representation of some aspect of the impact process are the two main constraints guiding the choice of one or the other approach. The ideal case would be an analytical model that gives results in good agreement with the numerical simulation. Unfortunately the complex and changing geometry during an impact makes simple numerical models impossible. Numerical simulations are then a necessary support to analytical models.

The first semianalytical energy balance model was developed by Gault and Heitowit (1963), who assumed that exactly half of the projectile's initial kinetic energy is trans-

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ferred to the target, uniformly distributed throughout a hemisphere of radius r enclosed by the expanding shock wave. Gault and Heitowit recognized two regimes of shock-wave pressure decay: A weak-shock regime in which pressure decay is proportional to $d^{-3/2}$, where d is the distance from the impact, and a strong-shock regime, where the pressure is proportional to the shock velocity squared, in which the pressure declines as d^{-3}. Melt and vapor regions are hemispheres whose radius corresponds to the distance from the impact point at which the shock pressure is equal to the melting or vaporization pressure. The Gault–Heitowit model requires the melt mass to be nearly proportional to the momentum of the impactor. This model, however, even after being refined with the introduction of the concept of the waste heat, underestimates the amount of melt and vapor production in an impact.

More recent semianalytical models have been developed by Croft (1982) and Tonks and Melosh (1993). In Croft's model, called the Gamma model, beyond a certain distance d from the impact point the shock pressure decay is

$$P(d) = P_0 \left(\frac{r_0}{d}\right)^n$$

where P_0 and r_0 are, respectively, the shock pressure and the radius of the *isobaric core*, a region where the pressure is constant or slowly decaying (Ahrens and O'Keefe name it "impedance match regime"). In this model the constant n must be either measured by experiments or determined by code calculations. The most recent determination of n is given in Ahrens and O'Keefe (1987) as a function of the impact velocity U (in km/sec): $n \cong -0.625 \log(U) - 1.25$ (Eq. 9 of their paper). Tonks and Melosh's model follows the same procedure as Croft's model but uses the particle velocity decay instead of the pressure decay.

Both semianalytical models make some important assumptions: the isobaric core is a sphere tangent to the surface; regions of melting and vaporization are truncated spheres; the centers of melt/vapor regions coincide with the center of the isobaric core.

We have carried out a detailed numerical study of the shock pressure and particle velocity decay regions; this allowed us to determine the exact location and shape of the isobaric core and of the melting and vaporization regions. The purpose is to test the assumptions made and to provide better estimates of the parameters needed in the semianalytical models.

The ultimate purpose of the semianalytical models is to estimate the amount of melt and vapor produced in impact events. Bjorkman and Holsapple (1987) developed, in the point source limit approximation ($R_{pr} \to 0$, $U \to \infty$), a scaling law for the amount of melt produced. Such a scaling law would eliminate the need for repeated calculations by relating the results of a hydrocode study to the impact event of interest. The scaling law when the projectile and target are composed of the same material is

$$\frac{V_M}{R_{pr}^3} \propto \left(\frac{E_M}{U^2}\right)^{-3\mu/2}, \tag{1}$$

where V_M is the volume of melt, R_{pr} is the projectile radius, U is the impact velocity, and E_M is the internal energy of melting.

Bjorkman and Holsapple's main concern was to establish whether the melt mass scaled with the momentum ($\mu = 1/3$) or the energy ($\mu = 2/3$) of the impactor. We applied this scaling law to the results of a series of hydrocode simulations with different impact velocities and materials. The result is a more accurate determination of the "velocity exponent" μ.

MODEL

Our simulations were done using the two-dimensional finite difference hydrocode CSQ III (Thompson 1988), complemented by the ANEOS equation of state package (Thompson and Lauson 1972), developed at Sandia National Laboratory. CSQ III is essentially a Eulerian code, in which the material moves through a fixed mesh. Unfortunately only vertical impacts can be modeled with the two-dimensional code. The most probable angle of impact is 45° which, however, can only be modeled with a three-dimensional code. We are hoping to be able, as soon as the code and the computer capabilities will become available to us, to extend this work to oblique impacts as well.

The effect of the mesh resolution in the determination of the amount of melt/vapor was studied by doing several runs at different resolutions and comparing the resulting melt/vapor produced. Figure 1 shows the results of the simulations for four different resolutions: 5, 10, 20, and 40 cells per projectile radius (cppr). The lowest resolutions, 5 and 10 cppr, are equal to the resolutions often used in previous works, while the highest resolution, 40 cppr, produced runs with cpu times already unreasonable (too long) for the many runs needed for the systematic character of this work. This test shows the clear deterioration in the determination of melt/vapor masses as the resolution decreases. The resolution effect is most dramatic for runs with lower impact velocities (low melt production), including most of the impact velocities of planetary interest ($v \leq 30$ km/sec), resulting in artificially small estimates of melt/vapor production by as much as a factor of 2. As a compromise between the need for an accurate melt/vapor determination and the need for relatively fast runs we have adopted a "20-cppr" resolution. This ensures a good resolution for our calculations without using excessive cpu

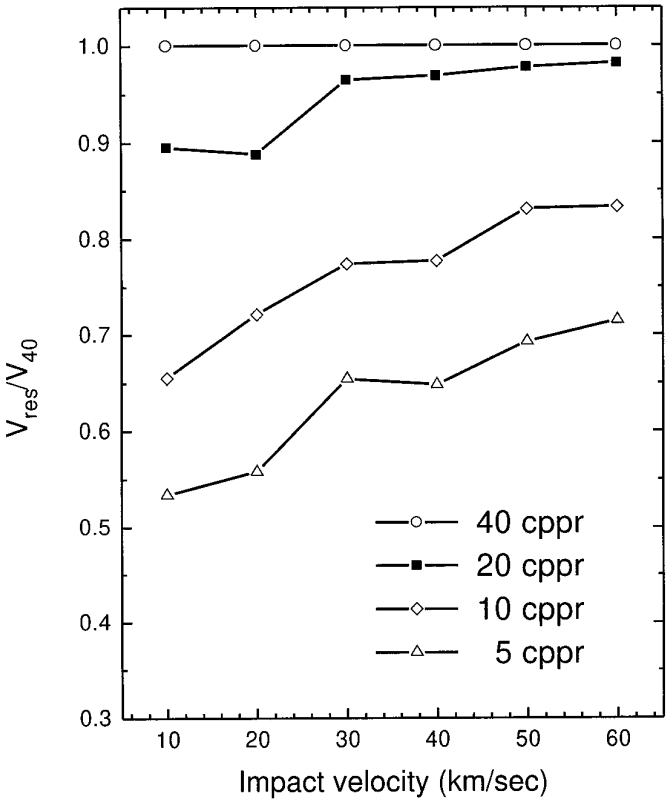

FIG. 1. Effect of the resolution on the melt volume estimates for various impact velocities. The lowest resolutions, 5 and 10 cppr, are similar to the resolutions used in previous work (e.g., O'Keefe and Ahrens 1977, Bjorkman and Holsapple 1987). Filled symbols represent the resolution used in this work: 20 cppr.

time. We have to keep in mind, however, that we may be underestimating the melt and vapor production in the lower velocity runs by as much as 10%.

The Eulerian coordinate system employed describes the instantaneous state of the material in each zone, but is unable to follow a point in the material through the mesh. Tracer particles are used for this purpose. The tracers are massless particles that move through the mesh using the local velocity at the current tracer position. They have no effect on the flow field but allow us to follow the thermodynamic history of some given material point in time.

The determination of melt and vapor volumes follows the method outlined in Pierazzo *et al.* (1995). Up to 100 tracer particles are used for the delineation of the melt/vapor regions in the target and projectile resulting from the impact. For our runs with a nominal projectile diameter of 4 km, Fig. 2 shows the typical disposition of the tracers around the impact point. The initial positions of the tracers allowed us to determine the volume of original target melted or vaporized. Previous work dealing with the calculation of the amount of melt and vapor produced in impact events used the shock pressure reached in the various cells at different times, introducing transport errors in the final melt and vapor volumes because it is difficult to correlate the shock pressure in a Eulerian zone with the material that is exposed to this pressure.

For a more refined calculation the entropy of partially melted/vaporized material was assumed to vary linearly with melt fraction; the Hugoniot was then used to determine the shock pressures associated with the partial melting/vaporization entropy. The melt and vapor regions were broken into shells, and the radius of each shell was computed by determining the distance at which the shock pressure is equal to the partial melt/vapor fraction. The total amount of melt and vapor was then determined by weighting the various shells' volumes by the corresponding melt/vapor fraction. This is particularly important for an accurate determination of the vapor produced in an impact event (complete vaporization shock pressures are reached only by the fastest events, while partial vaporization is reached in most cases).

THE EQUATION OF STATE

A high-speed impact causes a sudden compression of both projectile and target materials at the impact surface. A shock wave then propagates through both projectile and target, causing the material's thermodynamic state to

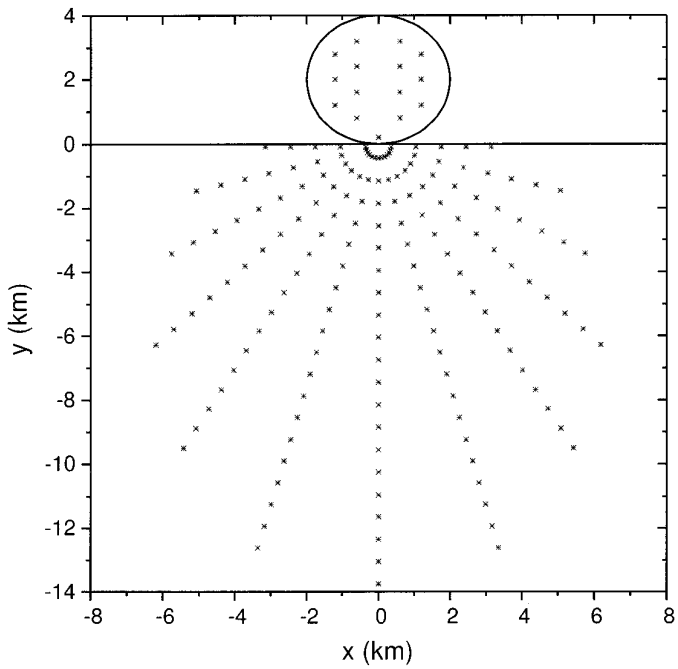

FIG. 2. Typical disposition of the Lagrangian tracer particles around the impact point. The spacing of the tracers varies according to the impact velocity, to cover a wider region as the impact velocity increases.

change discontinuously from its initial state to the shocked state. Specific material properties govern the response of materials to shocks, resulting in different behaviors for nominally the same impact conditions (same R_{proj} and U). An equation describing the thermodynamic state of a system over a wide range of pressures, temperatures, and specific volumes is necessary. This equation is called the equation of state (EoS) and it is unique for any material since it depends on the complex atomic, molecular, and crystalline structures of the materials.

The most widely used analytical EoS for impact studies is the Tillotson EoS, which was specifically derived for high-speed impact computations (Tillotson 1962). The Tillotson EoS is designed to fit the experimentally derived linear shock velocity–particle velocity relation at low pressures and to extrapolate to the Thomas–Fermi limit at high pressures. The analytical EoSs are limited in their representation of the thermodynamic state of a material over a wide range of pressures, temperatures, and densities. In particular, thermodynamic consistency might not be satisfied, especially when phase changes occur; changes in pressure and density in two-phase regions cannot be correctly described, resulting in the inability of the EoS to describe melting and vaporization.

More recent EoSs use increasingly complex computer codes that rely on different physical approximations in different domains of validity. In this work we use ANEOS (Thompson and Lauson 1972), a FORTRAN code designed for use with a number of hydrocodes (Thompson 1990). ANEOS was used also by Bjorkman and Holsapple (1987). Pressures, temperatures, and densities are derived from the Helmholtz free energy and are thus thermodynamically consistent. The properties of a given material are described in ANEOS by an array of 24 variables, which can be extended to 40 for a better description of mixed-phase material states, in addition to arrays containing the atomic number and the mass fractions of each element present in the material. The ANEOS library provides the EoS parameters for some simple materials; for most of the materials of geologic interest, however, the ANEOS EoS must be provided by the user.

The materials used in this work are granite, dunite, ice/water, aluminum, and iron. A library ANEOS EoS was available only for aluminum and iron, while for the other materials we provided the parameters for the EoS we developed (Benz *et al.* 1989 for dunite; Tonks *et al.* 1993 for ice/water; this work for granite). Figure A1 shows the ANEOS Hugoniot for granite compared with the experimental Hugoniot data.

The major limitation of ANEOS is illustrated in Fig. 3, which shows the thermodynamic path of shocked dunite in pressure–entropy space, where the entropy of complete vaporization is much higher than the experimental one and the critical point pressure is about 10 times higher than derived from the hard-sphere formalism (Ahrens and O'Keefe 1972). The reason for this is that the vapor entropy predicted by ANEOS includes the entropy of dissociation

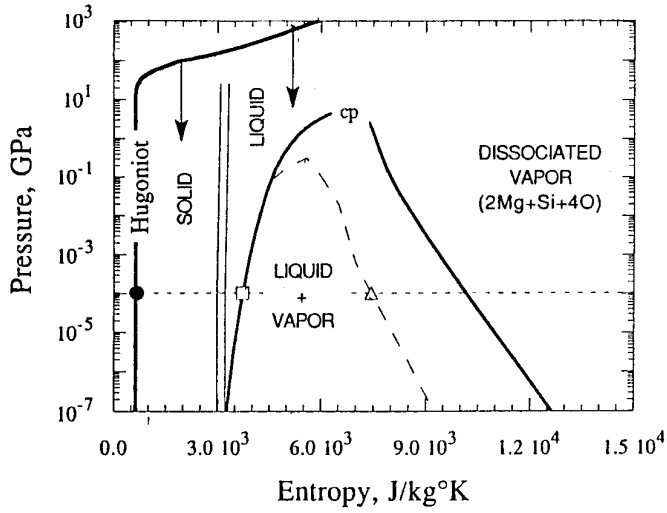

FIG. 3. Thermodynamic path of shocked dunite in pressure–entropy space. The Hugoniot curve and liquid–vapor phase curve are based on the ANEOS EoS (Benz *et al.* 1989). The two vertical lines between the solid and liquid phase, representing the 1-bar incipient and complete melting entropy, and the black circle, representing the entropy of the 1-bar reference state, are derived from the JANAF tables. The points of incipient (square) and complete (triangle) vaporization are from Ahrens and O'Keefe (1972). The long-dashed curve is an interpolation through the hard-sphere critical point (Ahrens and O'Keefe 1972), indicating the probable true nature of the liquid–vapor phase boundary. This boundary is well described by ANEOS on the liquid side, below the critical point (continuous line), but is badly represented on the vapor side (see text).

of dunite into the monoatomic gas mixtures (O, Si, Mg, etc.) instead of the molecular gas phases (mainly MgO, SiO_2, and O_2; Hashimoto *et al.* 1979). This limitation does not, however, affect the applicability of the EoS in the pressure range of interest.

The shock pressures required for incipient and complete melting and vaporization are given by the intersection of the Hugoniot with the corresponding melting/vaporization isentropes, represented, on a pressure-versus-entropy plot, by straight vertical lines. Values of the entropy of incipient and complete melting and vaporization at 1 bar were extracted from the literature (Chase *et al*, 1985; Ahrens and O'Keefe 1972, 1985) except for granite, for which a thermodynamic model was used (see the Appendix). Table I shows the relevant parameters for the materials used in this work; bulk sound speeds and densities where extracted from Melosh (1989).

RESULTS

Melting and vaporization in impact events are governed by the thermodynamics of shock compression and release. Shock compression is a thermodynamically irreversible process, for which the state of the shocked material jumps discontinuously from its initial state to a final state lying on the Hugoniot curve. The shock wave quickly reaches

TABLE I
Parameters Relevant to the Materials Used in This Work

	Dunite		Granite		Aluminum		Iron		Water/Ice	
	S^a (J/kg/K)	P^b (GPa)	S^c (J/kg/K)	P^b (GPa)	S^a (J/kg/K)	P^b (GPa)	S^a (J/kg/K)	P^b (GPa)	S^a (J/kg/K)	P^b (GPa)
Incipient melt.	3000	135	2220	46	2210	73	1650	310	2,210	0.4
Complete melt.	3240	149	2500	56	2640	106	1790	389	3,160	3
Incipient vapor.	3663[d]	186			3930	315	2240	887	4,950	4.5
Complete vapor.	7410[d]	2220			7830		4240		10,900	43
Melt. energy[b] (J/kg)	9×10^6		5.2×10^6		7.2×10^6		1×10^6		0.8×10^6	
Melt. enthalpy[b] (J/kg)	3.65×10^7		1.74×10^7		3.25×10^7		3.95×10^7		0.3×10^7	
C_p[e] (km/sec)	6.5		3.68		5.3		3.8		1.28	
ρ[e] (kg/m^3)	3320		2630		2700		7850		1000	

[a] JANAF Thermochemical Tables.
[b] ANEOS output.
[c] MELT Program.
[d] Ahrens and O'Keefe (1972).
[e] Melosh (1989).

the rear end of the projectile and is reflected back as a rarefaction wave that releases the compressed material to low pressures. In the absence of dissipative mechanisms (e.g., material rigidity, viscosity) the decompression of the material is an adiabatic (isentropic) process, i.e., thermodynamically reversible. This is the reason why we chose to represent the EoS in pressure–entropy space, since the release curve is a simple vertical line (Fig. 3).

Numerical simulations have shown that the maximum shock pressure generated by a high-speed impact has at least two distinct dependencies on distance from the impact site. Near the impact site the isobaric core delimits a region where the rate of pressure decay with distance is small. Analytical models typically assume the pressure to be constant within the isobaric core. Beyond this region is the "pressure decay regime," a region in which pressure as well as particle velocity decays rapidly with distance from the impact according to the power law

$$P(d) = A(d)^{-n},$$

where A is a constant. The semianalytical models need an estimate of the constant n for the determination of melt and/or vapor production. Ahrens and O'Keefe (1987) and Pierazzo *et al.* (1995) showed that the exponent n is not constant, but seems to depend on the impact velocity and the material type. A similar dependency is also true for the particle velocity, with the advantage that, in theory, the exponent for the particle velocity does not depend on the EoS, i.e., on the material, and follows a more general law of decline with impact velocity (Melosh 1989).

Figure 4 shows the decrease of the shock pressure with distance from the impact point, along the vertical symmetry axis, for various impact velocities, U. Each point corresponds to a tracer particle and records the maximum shock pressure experienced by it. The abrupt change in slope indicates the edge of the isobaric core and the beginning of the pressure decay regime.

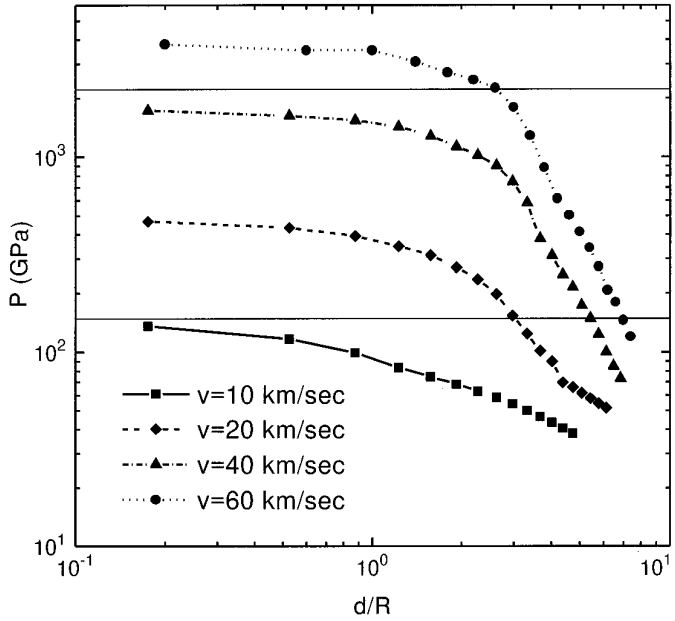

FIG. 4. Shock pressure decay from the impact point, along the vertical, for various impact velocities (dunite on dunite). Shock pressures corresponding to complete melting and vaporization for dunite are also shown as horizontal solid lines.

To better understand to what extent impact-related parameters can influence the exponent of the pressure decay law, we tested its dependence on various impact-related parameters: material type, impact velocity, projectile size, and angle from the vertical. On a log–log plot the power law in the pressure decay regime can be approximated by a straight line, whose slope is n:

$$\log(P_{\mathrm{sh}}) = a - n \log\left(\frac{d}{R_{\mathrm{pr}}}\right).$$

The distance from the impact point has been normalized to the projectile radius R_{pr} to eliminate the dependence on the size of the projectile. Runs were done with different projectile radii and the same set of impact velocities. A small dependence on the projectile size was observed for the shock pressure decay at the lowest impact velocities. We believe this is a consequence of the combination of finite resolution and smaller amount of melt produced (Fig. 1), since the fundamental equations of motion should be invariant under rescaling of distance and time, provided gravity- and rate-dependent processes are unimportant (Melosh 1989). A resolution higher than the 20 cppr used should solve the problem.

We find that the material type has a small influence on n; for analytical purposes, a general value for n would be desirable. In this work, a simple average of the values for all the materials but ice gives good results. Ice has an anomalously high compressibility due to its open lattice structure, making it more similar to a porous material. Values of the pressure decay exponent at angles of 0°, 15°, and 30° from the vertical (the least disturbed by surface effects and less affected by the angle dependence) were averaged for each run. A weighted least-squares fit of the average n as function of impact velocity was then carried out. The dependence of n on the impact velocity U, $n = a + b \log(U(\mathrm{km/sec}))$ (Ahrens and O'Keefe 1987), as seen in Figs. 5a and b for pressure decay and particle velocity decay, respectively, has given

$$a = -1.84 \pm 0.17, \quad b = 2.61 \pm 0.14,$$

for the shock pressure decay exponent, and

$$a = -0.31 \pm 0.09, \quad b = 1.15 \pm 0.06,$$

for the particle velocity decay. The correlation coefficients between the data points and the fitted lines are 0.948 and 0.943 for shock pressure decay and particle velocity decay, respectively. We conclude that the material dependency does not affect significantly the decay of the shock pressure compared with the material-independent particle velocity decay.

The chosen distribution of tracer particles allowed us to check the dependence of the power-law exponent on the angle from the vertical. Neglecting the shallowest angles (75°, 90°) where the surface significantly affects the shock pressure decay, a small angle dependence has been found; i.e., the power law exponent tends to decrease at shallower angles, as if the shock pressure contours are more compressed directly underneath the projectile. The effect is more pronounced at higher impact velocities. Figure 6 shows the slope of the angle dependence fit, $n = a + b\theta$ (where θ is in degrees) averaged for the various materials, versus the impact velocity.

ISOBARIC CORE

Analytical models usually assume a spherical shape for the isobaric core. We estimated the size of the isobaric core by determining the point where the two pressure decay regimes intersect (Fig. 4). The two regimes where determined by fitting a line through the two tracer points closest to the impact point for the first regime and by a least-squares fit of a number of tracer points for the second (avoiding the intermediate region, where the shift from one regime to the other occurs). The shape of the isobaric core in analytical models has always been represented by a sphere. Figures 7a and b show the data points for the isobaric core (represented with a ×) and the least-squares interpolation using a circle. Excluding the near-surface data points, i.e., avoiding the near-surface interference zone where the pressure decrease is much steeper converging to zero at free surface, the interpolation by a circle seems reasonable.

A small dependence of the size of the isobaric core (relative to the projectile radius) on the impact velocity is found. Figures 8a and b show, respectively, the isobaric core radius and center depth as a function of the logarithm of the impact velocity for the various materials used. A least-squares fit of the data to a straight line, $\log(r_{\mathrm{ic}}/R_{\mathrm{proj}}) = a + b \log(U)$, gives

$$a = -0.346 \pm 0.034, \quad b = 0.211 \pm 0.022.$$

The same procedure for the center depth of the isobaric core ($d_{\mathrm{ic}}/R_{\mathrm{pr}}$) gives

$$a = -0.516 \pm 0.060, \quad b = 0.361 \pm 0.038.$$

Note that, as the impact velocity increases, the depth of burial increases faster than the radius, in contrast to previous analytical models which assumed the top surface of the isobaric core to be tangent to the free surface. This suggests that as the impact velocity increases, the projectile releases the bulk of the energy deeper in the target.

A similar conclusion is reached if the particle velocity

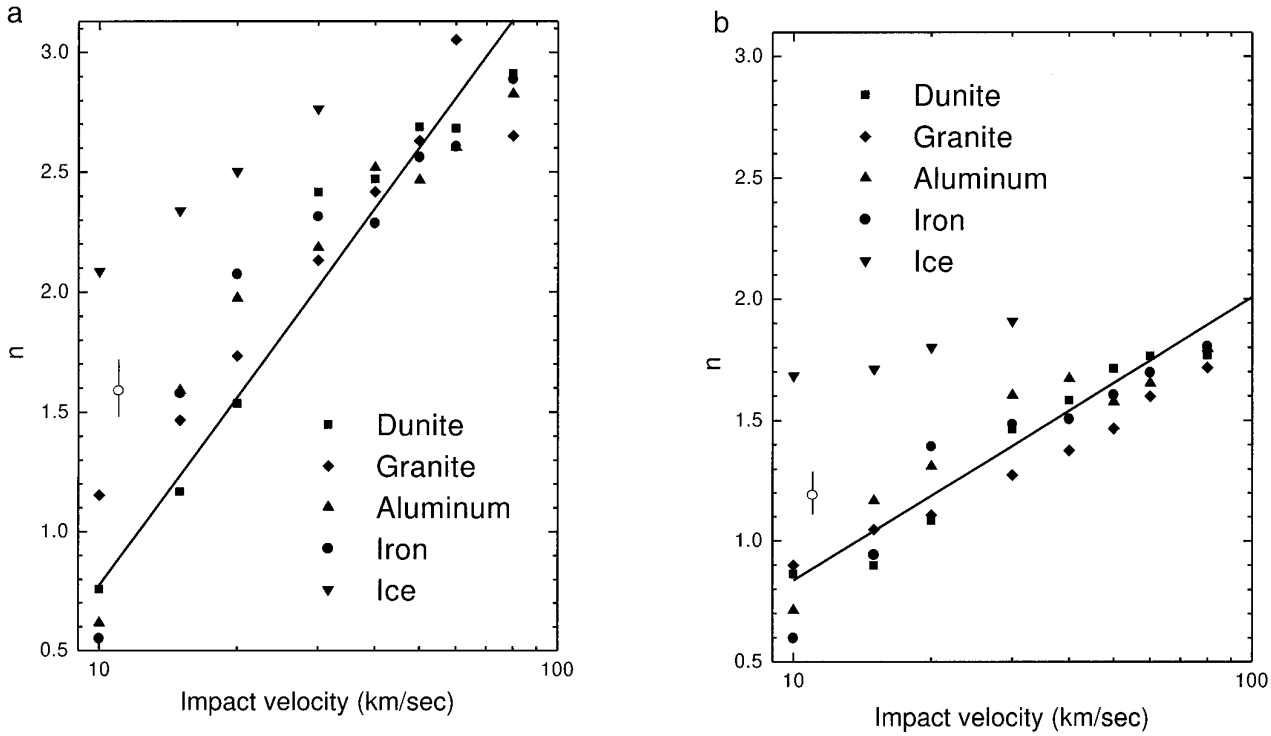

FIG. 5. Power-law exponent versus impact velocity for (a) shock pressure decay, and (b) particle velocity decay. The heavy solid line represents the best fit (weighted least squares) of all the data points but ice. Also indicated as a vertical bar centered on an open dot is the average uncertainty for the data points.

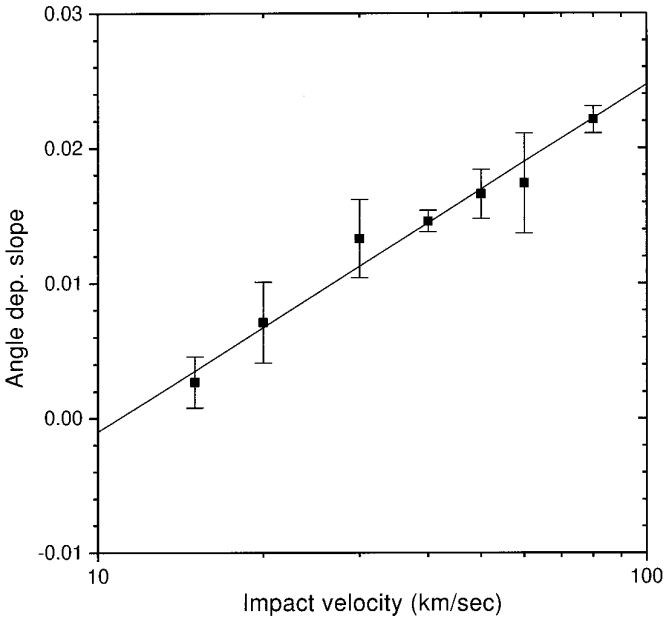

FIG. 6. Slope of the angle dependence, $n = a + b\theta$, of the power-law exponent, averaged for the various materials, as function of the impact velocity.

decay with distance from the impact point is used to define the isobaric core. In Table II the least-squares fits for each material separately and for all the materials but ice are given together, for both shock pressure decay and particle velocity decay.

MELT AND VAPORIZATION REGIONS

Previous semianalytical models have assumed shapes of melt and vapor regions varying from a hemisphere to a truncated sphere. Numerical calculations by Vickery and Melosh (unpublished results, 1992) and by Pierazzo *et al.* (1995) indicate that the melt and vapor regions are really squashed spheres. Using an analytical model, Tonks and Melosh (1993) determined the amount of melt estimated with both the hemispheric and the truncated sphere models. The two estimates differ, depending on the size of the event, from a factor of 5 for Brent-size events (terrestrial crater, 3.8 km in diameter) to a factor of 2 for Popigai-size events (terrestrial crater, 100 km in diameter) (Fig. 9 of their paper). A good estimate of the shape of the melt/vapor regions is therefore important for a more accurate estimate of the amount of melt and vapor produced in impact events.

We determined the shape of the melt and vapor regions

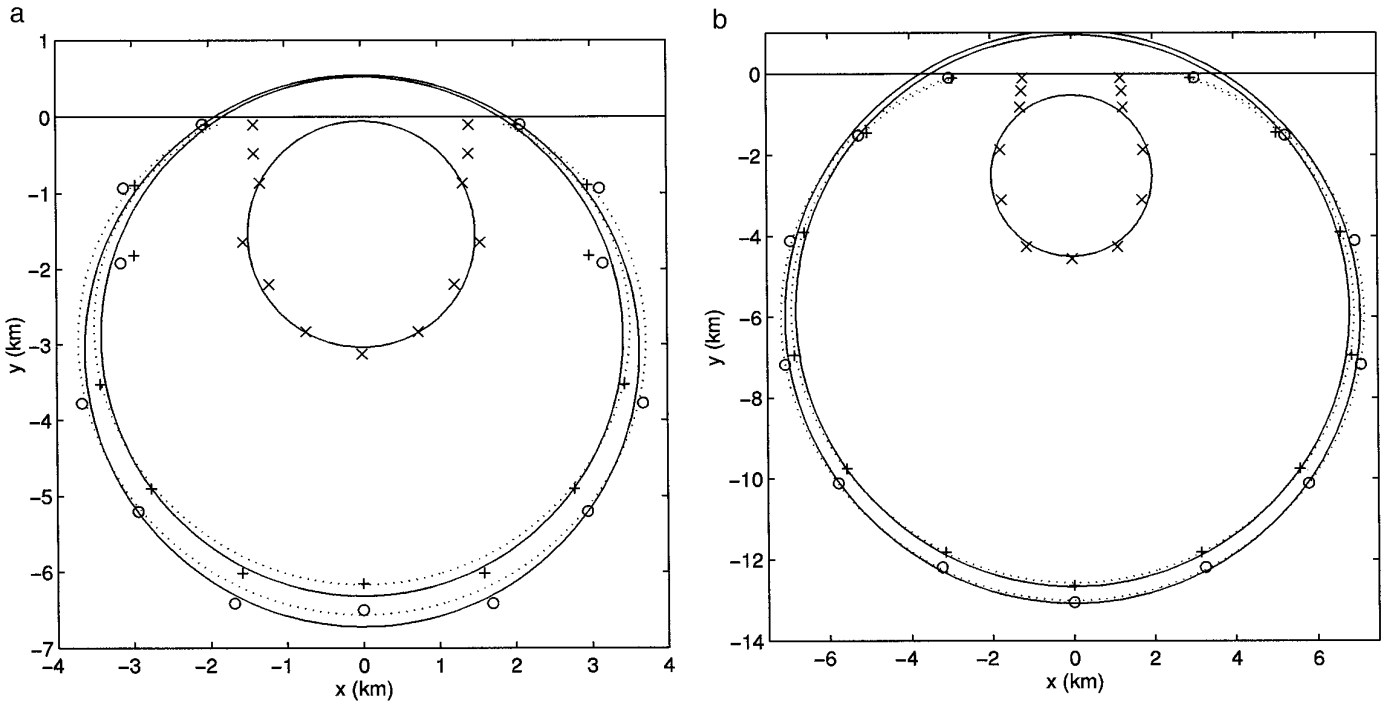

FIG. 7. Melting regions and isobaric core for (a) $U = 20$ km/sec and (b) $U = 50$ km/sec. Continuous lines: best fit of the data with a circle; dotted lines: best fit of incipient and complete melting data with a limacon of Pascal. ○, incipient melting; +, complete melting; ×, isobaric core data.

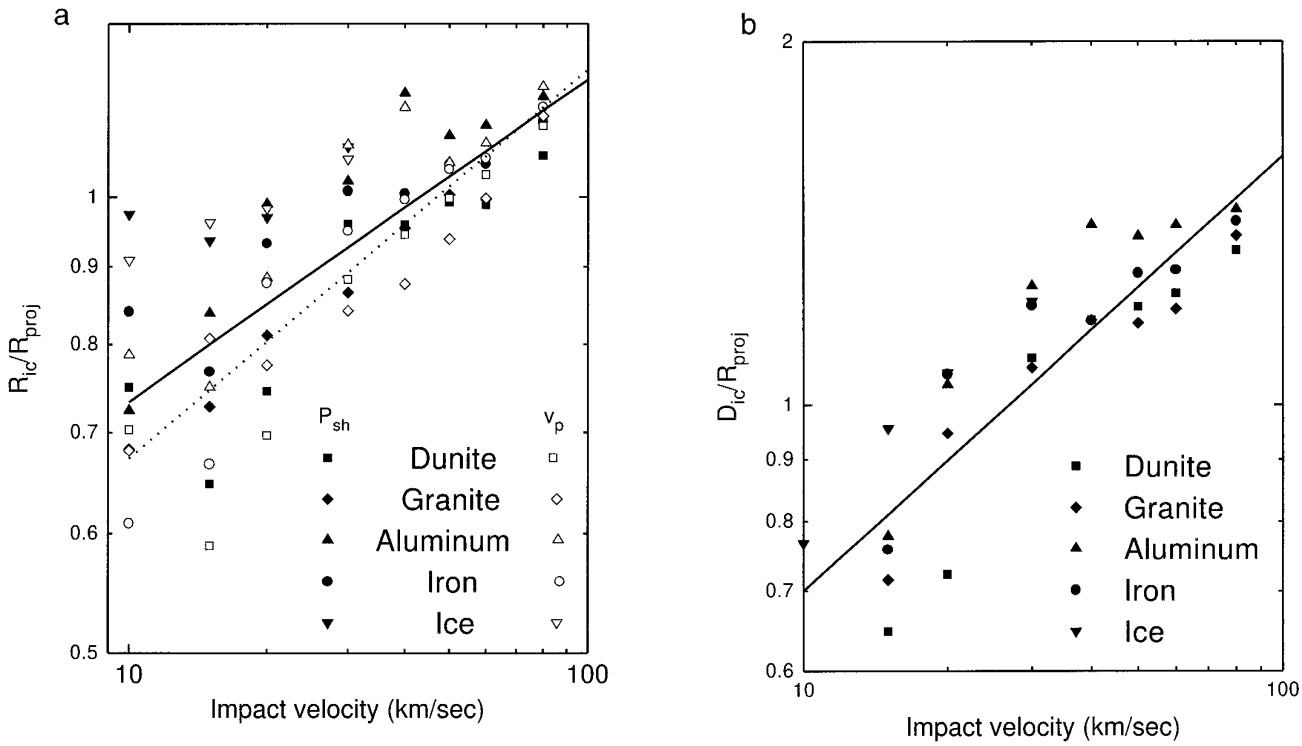

FIG. 8. Isobaric core: (a) radius and (b) depth versus impact velocity. (a) Heavy solid line and filled symbols indicate pressure decay data; dotted line and open symbols indicate particle velocity decay data. (b) Data and interpolation for pressure decay.

TABLE II
Dependence of Radius and Depth of the Isobaric Core on the Impact Velocity, According to the Shock Pressure and Particle Velocity Decay, Respectively

	$\log(r_{ic}/R_{pr}) = a + b \log(U)$			$\log(d_{ic}/R_{pr}) = a + b \log(U)$		
	a	b	Correlation	a	b	Correlation
			Shock Pressure			
All material (no ice)	−0.346 ± 0.034	0.211 ± 0.022	0.866	−0.516 ± 0.060	0.361 ± 0.038	0.880
Dunite	−0.391 ± 0.069	0.022 ± 0.012	0.894	−0.69 ± 0.11	0.451 ± 0.069	0.947
Granite	−0.409 ± 0.019	0.238 ± 0.012	0.992	−0.493 ± 0.082	0.335 ± 0.052	0.946
Aluminum	−0.327 ± 0.055	0.220 ± 0.036	0.927	−0.46 ± 0.11	0.347 ± 0.071	0.909
Iron	−0.256 ± 0.045	0.161 ± 0.029	0.913	−0.42 ± 0.10	0.360 ± 0.065	0.904
Ice	−0.085 ± 0.061	0.061 ± 0.049	0.663	−0.174 ± 0.067	0.374 ± 0.054	0.980
			Particle Velocity			
All material (no ice)	−0.428 ± 0.037	0.256 ± 0.024	0.889	−0.606 ± 0.080	0.408 ± 0.051	0.845
Dunite	−0.504 ± 0.075	0.291 ± 0.049	0.926	−0.94 ± 0.10	0.587 ± 0.064	0.972
Granite	−0.372 ± 0.038	0.209 ± 0.025	0.959	−0.350 ± 0.039	0.239 ± 0.025	0.974
Aluminum	−0.338 ± 0.063	0.221 ± 0.041	0.909	−0.459 ± 0.096	0.343 ± 0.061	0.929
Iron	−0.499 ± 0.055	0.303 ± 0.036	0.960	−0.67 ± 0.18	0.46 ± 0.11	0.876
Ice	−0.112 ± 0.026	0.084 ± 0.022	0.909	−0.14 ± 0.14	0.14 ± 0.13	0.533

by plotting the distance from the impact point at which the critical pressures for melting and vaporization are reached. Figures 7a and b show the points delimiting the incipient (circles) and complete (crosses) melt regions for two impact velocities, 20 and 50 km/sec, respectively; a continuous line represents the least-squares interpolation by a circle. An analytical expression of the "squashed sphere" described by Vickery and Melosh can be conveniently provided by a Limacon of Pascal equation ($r = a + b \cos \theta$, dotted line), which we compare with the simpler spherical shape previously used in analytical models. Except for the near-surface interference region, the two curves are always quite close and the amounts of melt and vapor obtained using the two models are both very close to the hydrocode-derived ones. Figure 9 shows the deviations from the hydrocode-derived melt volumes for the spherical model; all the deviations are within 10%, very similar to the deviations obtained with the Limacon of Pascal shape. When the melt/vapor region is small compared with the projectile size (particularly true for low impact velocities), neither analytical shape fits the computed points very well. In this case the melt/vapor regions are contained mostly in the near-surface interference zone. Overall we can conclude that even though the melt/vapor regions look like squashed spheres (due to the near-surface effects), the truncated sphere model for the melt/vapor regions is still a valid approximation for the melt and vapor regions.

As expected, the size of the melt/vapor regions depends strongly on the impact velocity and on the material type. Figure 10a shows a log–log plot of the radius of the complete melting region versus the impact velocity for all the materials. A least-squares fit for each material separately,

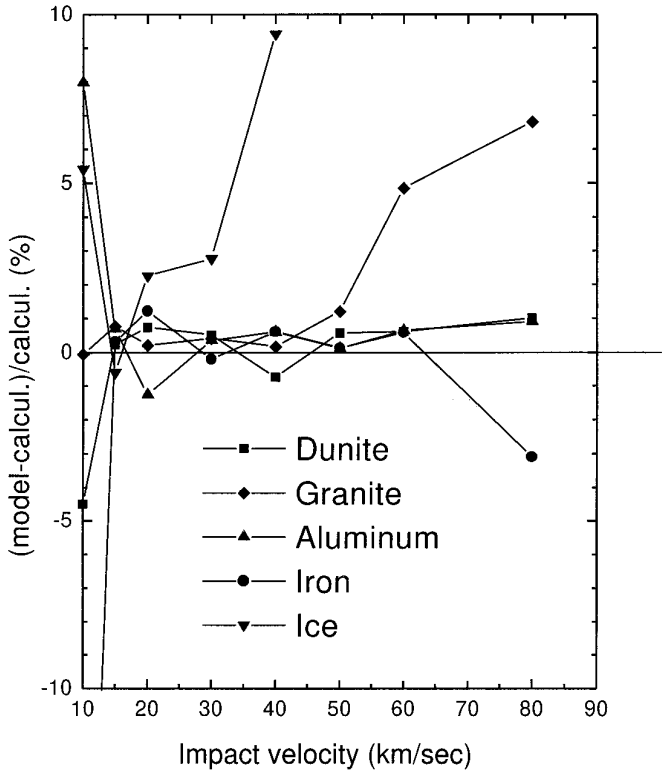

FIG. 9. Deviations, in percent, of the melt volumes derived from the spherical model (interpolation of the melt regions with a circle in two dimensions) from the hydrocode-derived ones (see text). The highest deviations occur for $U = 10$ km/sec, as expected.

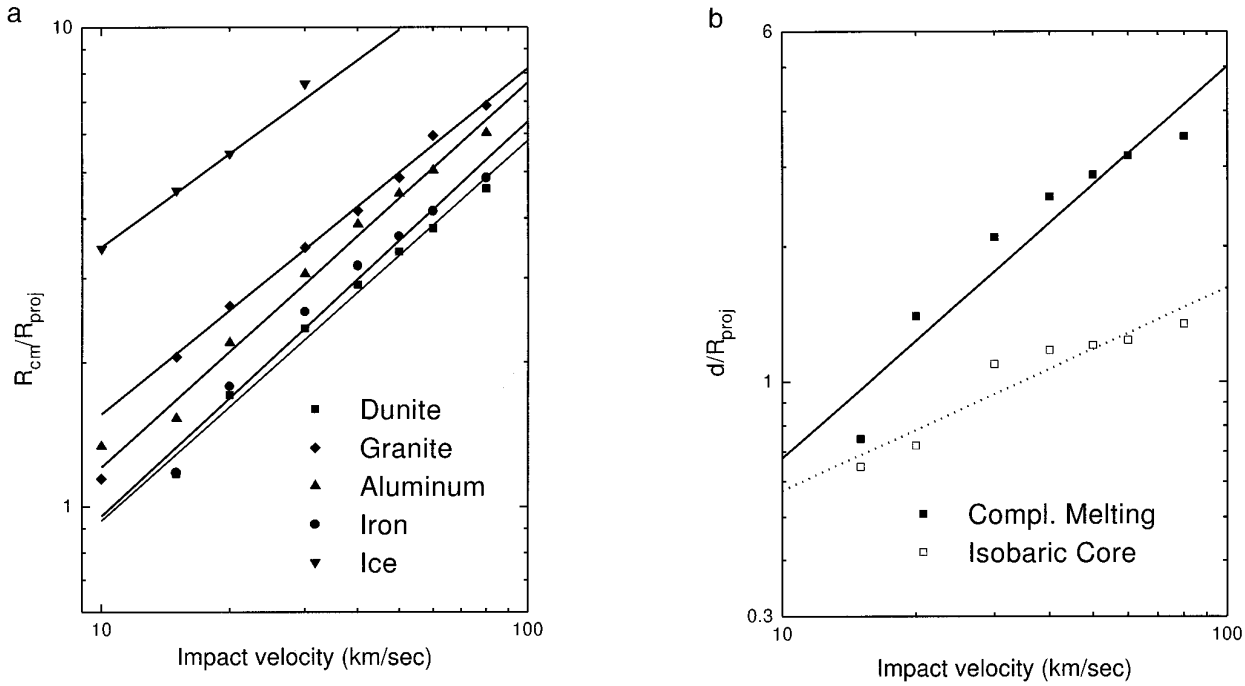

FIG. 10. (a) Radius of complete melting region (normalized to the projectile radius) versus impact velocity. Continuous lines represent the least squares fit for the various materials (as summarized in Table III). $U = 10$ km/sec has been excluded from the interpolations. (b) Data and the least squares fit of the complete melting (filled squares, solid line) and the depth of the isobaric core (open squares, dotted line) for dunite.

summarized in Table III, shows a similar slope that represents the dependence on the impact velocity, while the intercept, varying for the various materials, represents the material dependence. We excluded from the interpolation the case with $U = 10$ km/sec because the melt region is highly distorted by near-surface effects.

In Fig. 10b the least-squares fit of the center depth of the complete melting region for dunite versus impact velocity U is compared with the equivalent general interpolation depth of the isobaric core (see Tables II and III). It is evident that the assumption that the centers of the melt and vapor regions coincide with the center of the isobaric core (Croft 1982, Tonks and Melosh 1993) is incorrect, and can result in inaccurate melt and vapor volume estimates.

MELT PRODUCTION

The lack of experimental data on melt production (laboratory experiments cannot reach the impact velocities necessary for a significant production of melt) seriously hampers the efforts to constrain melt and vapor production models to ensure accuracy. Only a few estimates of melt production in well-studied terrestrial craters are available to test the estimates from theoretical studies and computations.

An important approach to melt and vapor production in impact events comes from the coupling parameter theory which is directly derived from the late-stage equivalent concept (Dienes and Walsh 1970). This approach produced

TABLE III
Dependence of Radius and Depth of the Complete Melting Region on the Impact Velocity

	$\log(r_{cm}/R_{pr}) = a + b\log(U)$			$\log(d_{cm}/R_{pr}) = a + b\log(U)$		
	a	b	Correlation	a	b	Correlation
Dunite	−0.823 ± 0.067	0.793 ± 0.043	0.993	−1.04 ± 0.19	0.87 ± 0.12	0.956
Granite	−0.527 ± 0.031	0.720 ± 0.019	0.998	−0.07 ± 0.12	0.34 ± 0.07	0.895
Aluminum	−0.721 ± 0.067	0.803 ± 0.043	0.993	−0.75 ± 0.17	0.751 ± 0.11	0.950
Iron	−0.844 ± 0.091	0.834 ± 0.058	0.988	−1.10 ± 0.25	0.92 ± 0.16	0.933
Ice	−0.104 ± 0.013	0.646 ± 0.010	0.999	−0.052 ± 0.059	0.476 ± 0.047	0.990

the scaling law proposed by Bjorkman and Holsapple (1987), Eq. (1). Following Bjorkman and Holsapple (1987) we used, for E_M, the specific energy of the Rankine–Hugoniot state from which the isentropic release ends at the 1-bar point on the liquidus. The coupling parameter theory predicts that the velocity exponent μ varies between 0.55 and 0.6, inbetween energy scaling and momentum scaling, for all nonporous materials. As the minimum impact velocity for melt or vapor production is approached, however, the melt or vapor is produced in a small region immediately adjacent to the impactor, and the shape and volume of the region are thus sensitive to factors like the shape of the impactor and the nearness to the free surface. In this case the point source limit is not applicable. Additionally, for impacts where the projectile and the target are made of different materials a factor describing the density contrast must be introduced in the scaling law.

In their study of melt production, Bjorkman and Holsapple (1987) carried out a series of hydrocode simulations with impact velocities varying from 5 to 100 km/sec. Their conclusion was that for $U^2/E_M < 50$ the total volume of melt scaled with the energy of the impact ($\mu \sim 0.66$), while above 50 the scaling was inbetween the energy and momentum ($\mu < 0.66$), in better agreement with their model prediction. The energy scaling of the melt volume is, however, not a new result. Previously O'Keefe and Ahrens (1977), after carrying out a series of hydrocode calculations, concluded that above a certain limit the total volume of melt and vapor produced in an impact event scaled as the energy of the impactor.

We applied the point source limit theory to our hydrocode simulations. Figure 11 shows the total amount of melt + vapor (corresponding to the region shocked above melting) as a function of the melt number U^2/E_M. In log–log form the expression is

$$\log\left(\frac{V_M}{V_{pr}}\right) = a + \frac{3}{2}\mu \log\left(\frac{U^2}{E_M}\right),$$

and a simple least-squares fit is required to determine the velocity exponent that best fit the hydrocode data. The breakdown of the point source limit as the impact velocity approaches the threshold for melt production is evident for $U^2/E_M = 30$. The results of the interpolation for the various materials are reported in Table IV. Since the material dependence seems negligible for the materials used in this work, with the exception of the ice-on-ice case, a general least-squares fit of all the hydrocode data for all the materials but ice was done. The result of the general interpolation,

$$\mu = 0.708 \pm 0.039,$$

is in good agreement with the energy scaling values of 2/3. The same interpolation applied to the ice-on-ice cases gives $\mu = 0.432 \pm 0.036$. This value is in agreement with the value of $\mu = 0.4$ proposed by Holsapple and Schmidt (1987) for "perfectly porous" materials. We see no evidence of the "less than energy" scaling for $U \geq 50$ km/sec found by Bjorkman and Holsapple (corresponding to only two data points in their work). Applying this cutoff to our results, we find $\mu = 0.69 \pm 0.14$. Figure 11 also shows, for comparison, the hydrocode results of Bjorkman and Holsapple (1987) for aluminum-on-aluminum, represented by ×, and of O'Keefe and Ahrens (1977) for anorthosite-on-anorthosite, represented by +. Our least-squares fit of their data is also reported in Table IV. We believe that the smaller amount of melt + vapor for these two previous works is due to the lower resolution (≅10 cppr) used in the hydrocode simulations, as compared to ours (see Fig. 1, open diamonds).

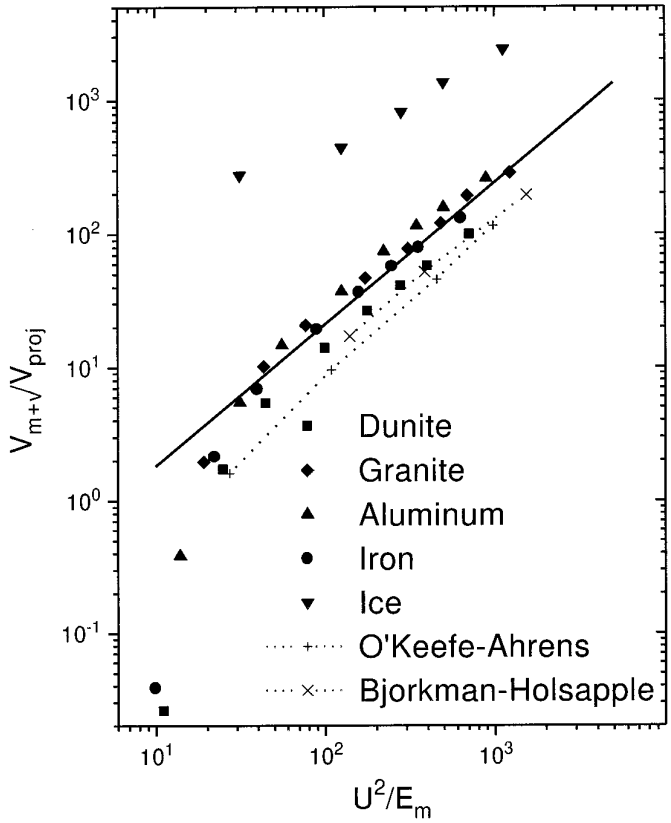

FIG. 11. Melt + vapor volumes versus melt number, U^2/E_M, for the various materials used in the simulations. The heavy solid line represents the best fit of all the data but ice, for $U^2/E_M > 30$. Below 30 the point source limit clearly breaks down. Ice data, shown as open triangles, do not fall in the same region as the other materials, but are characterized by a similar slope. Data from O'Keefe and Ahrens (1977) for anorthosite (+) and Biorkman and Holsapple (1987) for aluminum (×), connected by dotted lines, are also shown for comparison.

TABLE IV
Interpolation of Hydrocode-Generated Melt + Vapor Volumes for Various Impact Velocities

	$\log(V_{m+v}/V_{pr}) = a + \frac{3}{2}\mu \log(U^2/E_M)$		$U^2/E_M > 30$
	a	μ	Correlation
All data (no ice)	−0.80 ± 0.14	0.709 ± 0.041	0.964
Dunite	−0.972 ± 0.064	0.701 ± 0.018	0.998
Granite	−0.595 ± 0.064	0.667 ± 0.017	0.998
Aluminum	−0.636 ± 0.095	0.699 ± 0.026	0.997
Iron	−0.811 ± 0.099	0.708 ± 0.029	0.997
Ice	1.36 ± 0.13	0.432 ± 0.36	0.99
Bjorkman–Halsapple	−1.39 ± 0.16	0.752 ± 0.046	0.994
O'Keefe–Ahrens	−0.93 ± 0.10	0.676 ± 0.025	0.999

According to these results the point source limit predictions cannot be applied to planetary-type impacts ($U \leq 50$ km/sec), while the energy-scaling seems to be the more appropriate scaling law. This is not surprising since the late stage equivalence concept (on which the point-source scaling law is based) was derived for flow fields, not for shocks, which are directly responsible for the production of melt and vapor in impact cratering events.

COMPARISON WITH TERRESTRIAL CRATERS

To test the hydrocode estimates we compared our results with the terrestrial crater data set proposed by Grieve and Cintala (1992). They constructed a list of terrestrial craters whose impact melt volumes had been previously estimated from geological studies; the degree of crater erosion plus uncertainties in the method contribute to the large scatter in the data set. Table V lists the craters and relevant data (see also Table I and the appendix of their paper for more information). The transient crater diameter, D_{tc}, was calculated using Croft's (1985) scaling relation for complex terrestrial craters (except for Brent, which is a simple crater: $D_{tc} \cong D_{rim}$),

$$D_{tc} \cong D_Q^{0.15\pm0.04} \cdot D_{rim}^{0.85\pm0.04},$$

where D_Q is the simple-to-complex transition diameter which, for crystalline targets on Earth, is 4 km, and D_r is the rim-crest diameter of the crater.

Since we did not carry our computations out to times long enough to determine transient crater diameters, given the long computational time required, we calculated the transient crater diameter to associate to the hydrocode impact simulations using the Schmidt and Housen (1987) scaling relation,

$$D_{tc} = 1.16 \left(\frac{\rho_p}{\rho_t}\right)^{1/3} D_{pr}^{0.78} U^{0.43} g^{-0.22},$$

where ρ_p and ρ_t are the projectile and target density, D_{pr} is the project diameter, U is the projectile impact velocity, and g is the gravitational acceleration (g = 980 cm/sec).

We present the results for a dunite projectile impacting a dunite target. The impact velocities are 20 and 40 km/sec, while the diameter of the projectile varies from 0.2 to 10 km. Table VI lists the hydrocode results, which are compared to the terrestrial crater data in Fig. 12. While Grieve and Cintala's (1992) model results seem to overestimate the amount of melt produced, the agreement between our hydrocode results and the terrestrial data is very good. It must be kept in mind that the data set has been reconstructed by carefully choosing only craters in crystalline

TABLE V
Estimated Original Impact Melt Volumes within Terrestrial Craters as from Grieve and Cintala (1992)

Crater	D_{rim} (km)	D_{tc} (km)	V_{melt} (km^3)	
Brent	3.8	3.8	0.02	(fairly reliable)
Zapadnaya	4.5	4.42 ± 0.02	0.1	(minimum estimate)
Il'inets	8	7.21 ± 0.20	0.7	(minimum estimate)
Kahuga	15	12.3 ± 0.6	8	(could be 8.5)
Logoisk	17	13.7 ± 0.8	0.07	(fairly reliable)
Lappajärvi	17	13.7 ± 0.8	8	(quite uncertain; minimum estimate)
Ries	24	18.3 ± 1.3	0.2	(up to 4–5, Graup 1984)
Boltysh	25	19 ± 1.4	11	(fairly reliable)
Mistasin	28	20.9 ± 1.6	20	(uncertain by at least a factor of 2)
W. Clearwater	35	25.3 ± 2.2	80	(uncertain by at least a factor of 2)
Kara	65	42.8 ± 4.8	480	(minimum estimate)
Manicouagan	100	61.7 ± 8.0	1200	(minimum estimate)
Popigai	100	61.7 ± 8.0	1750	(fairly reliable; minimum estimate)
Sudbury	200	111.2 ± 17.5	8000	(quite uncertain; minimum estimate)

basements; when the amount of sediments in the target is nonnegligible, the presence of volatile materials (e.g., CO_2 and H_2O) greatly decreases the amount of melt produced (see Kiefer and Simonds 1980). This effect is illustrated by two craters in Table V, Ries and Logoisk, that in Fig. 12 are represented by open diamonds. To the present day, no good model for melt production for impact craters in sedimentary targets has been developed.

SUMMARY

A series of hydrocode calculations has been performed for vertical impacts. Impact velocity as well as projectile radius and material type has been varied in the simulations. The result is a data set that can be used as a basis for studies and models of impact cratering events.

We first checked the assumptions normally made in semianalytical models. We defined the shape of the isobaric core and melt/vapor regions and obtained the dependence of radius and depth of such regions on impact velocity. We reached six major conclusions: (1) The isobaric core does not remain always tangent to the surface, as assumed by Croft (1982), and Tonks and Melosh (1993), but tends to sink below the surface as the impact velocity increases (Fig. 7). (2). While the melt/vapor regions increase significantly in size with increasing impact velocity ($\propto U^{0.65-0.84}$), the isobaric core radius relative to impactor size shows only a small dependence on impact velocity ($\propto U^{0.06-0.21}$). (3) Near-surface effects are responsible for altering the shape of the isobaric core and melt/vapor regions from ideal shapes (Fig. 7). (4) Even though melt/vapor regions look like "squashed spheres," the assumption of a spherical shape is still a good approximation and can be used in analytical models (Fig. 10). (5) In disagreement with the analytical models' assumption, the centers of melt/vapor regions do not coincide with the center of the isobaric core but they tend to move further down into the target as the impact velocity increases (Fig. 8b). (6) For crystalline materials, the shock pressure decay with distance from the impact point is not a strong function of the material type, as previously thought (Fig. 5).

The quantity of melt and vapor produced in an impact influences various aspects of the impact cratering event, from heat deposition, to the development and composition of the vapor plume, to the crater shape and impact lithology. Unfortunately, due to the impossibility of performing sizable high-speed laboratory impact experiments, no experimental data on substantial melt production are available at this time, and only a few observational data, in many instances still controversial, are available to constrain and test the theoretical results of models and/or computer simulations.

We determined the amount of melt and vapor produced in our impact cratering hydrocode simulations, and compared our results with the point source limit scaling law proposed by Bjorkman and Holsapple (1987). Our results clearly support energy scaling for the melt volume, as inferred by Ahrens and O'Keefe (1977). While Bjorkman and Holsapple conclude that the point source limit scaling ($\mu \sim 0.55$–0.6 for nonporous materials) is satisfied by the results of their hydrocode simulations for $U \geq 50$ km/sec, we believe that the low resolution and limited number of

TABLE VI

Hydrocode Melt + Vapor Volumes for Dunite Impacting on Dunite for Different Projectile Diameters and Impact Velocity[a]

	U = 20 km/sec		U = 40 km/sec	
D_{pr} (km)	D_{tc} (km)	V_m (km^3)	D_{tc} (km)	V_m (km^3)
0.2	3.42	0.019	4.64	0.09
0.5	7	0.26	9.48	1.31
1	12	2.49	16.28	10.72
2	20.61	19.19	27.96	83.28
3	28.28	67.36	38.36	290.83
4	35.39	163.03	48.01	698.45
6	48.56	583.56	65.88	2,451.04
10	72.33	2,935.19	98.12	10,918.08

[a] Transient crater diameter determined using Schmidt and Housen (1987) scaling relation.

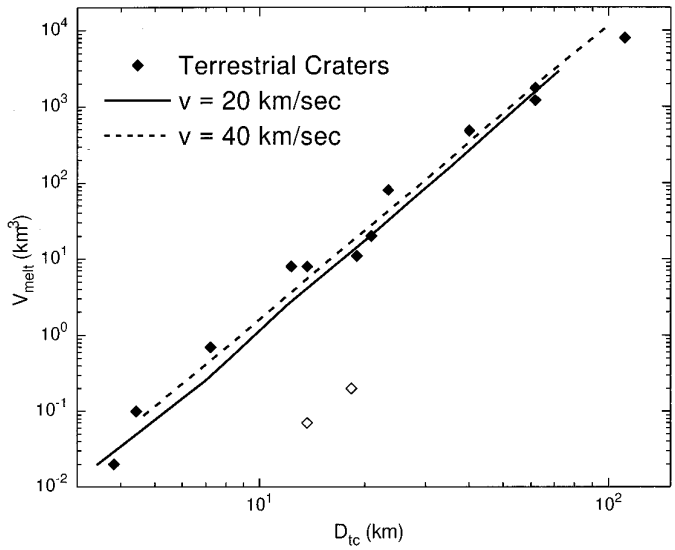

FIG. 12. Hydrocode-derived impact melt volumes versus transient crater diameter, compared with terrestrial data (diamonds). The simulations are for a dunite projectile impacting a dunite target at 20 and 40 km/sec. The lines connect hydrocode calculations corresponding to D_{pr} = 0.2, 0.5, 1, 2, 3, 4, 6, 10 km. Open diamonds indicate the Ries and Logoisk data, for which the thick sedimentary layer is thought to greatly affect the production of melt.

simulations (3 with the "very-fine resolution," corresponding to 10 cppr, of which only two had $U \geq 50$ km/sec) seriously weaken their conclusion. We conclude that, for planetary-type impact events where the impact velocities exceed about 15 km/sec, the amount of melt and vapor can be scaled by the energy of the impactor (Fig. 11, Table IV).

The only material that does not follow the general behavior found above is ice/water. In particular the amount of melt + vapor produced in planetary-type events on ice is about one order of magnitude higher than on other materials, as can be seen in Fig. 11. This might affect the final shape and appearance of the resulting craters and needs to be further investigated, given the importance of water and other ices in planetary surfaces of the outer Solar System.

Finally, to verify the accuracy of our melt production calculations, we tested our results against a data set of well-studied terrestrial craters (Grieve and Cintala 1992). Figure 12 shows good agreement between our theoretical calculations and the available observational data.

APPENDIX: THE ANEOS EQUATION OF STATE FOR GRANITE

Granite is an igneous rock consisting chiefly of feldspars and quartz, with a small amount of mica or hornblend. Clarke and Washington (1924) estimate that the average composition of the continental crust is intermediate between that of granite and basalt. Granite can thus be used to represent a typical upper continental crust material on the Earth, and its EoS is important for modeling terrestrial impact events.

The major difficulty in developing the EoS of granite lies in the fact that it is a complex mixture of minerals, each with its own thermodynamic properties. There is a complete series grading from granite, with mainly K-feldspar and no oligoclase, to granodiorite where the feldspar is mostly plagioclase and only slightly more than 5% K-feldspar. For any rock a single melting and vaporization temperature is a considerable simplification, since the phase diagram for a mixture will, in general, be quite complicated. Nevertheless, the average melting and vaporization temperatures of the major constituents of the mixture can be used to estimate the specific internal energy of the mixture in the phase change region. In the case of granite, we use $T_m = 1673$ K for the melting temperature, as given by Allen (1967), while we estimated the vaporization temperature, $T_v = 3600$ K, as a weighted average of the available vaporization temperatures of its components (Ahrens and O'Keefe 1972).

Among all the possible granite compositions available, we developed the ANEOS EoS for Westerly granite ($\rho = 2.63$ g/cm^3), for which numerous shock experimental data are available (e.g., Marsh 1980, McQueen *et al.* 1967, Birch 1966). Westerly granite, from Rhode Island, consists of 35.4% microcline, 31.4% $An_{20}Al_{80}$, 27.5% quartz, and 3.2% mica. The bulk composition, in mole percentages of oxide components (Govindaraju 1984), of Westerly granite was used to reconstruct the mass fraction of the seven most abundant elements used in ANEOS (Table AI).

Shock experiments performed on granite indicate that a high-pressure solid–solid phase transition occurs somewhere between about 5 and 10 GPa (McQueen *et al.* 1967). This is believed to be the result of a solid–solid phase transformation (low- to high-density phase) for quartz, which constitutes about one-third of the rock on a mass basis. The density increase in this type of transition is associated with compression work (PdV) that sensibly enhances the shock heating of the material, and on the Hugoniot it appears as an abrupt change in slope. Since the current version of ANEOS cannot deal with liquid–solid and solid–solid transitions simultaneously, we chose to favor the latter, which clearly affects the shock states of the material, and ignore the details of the melt region (low P). As a result, we could not use the melt transition phase at low P as a constraint in the building of the EoS for granite as we did for other materials, like water (Tonks *et al.* 1993). Figure A1 shows the excellent agreement of the Hugoniot computed by ANEOS with the experimental data, while the input parameters for the granite ANEOS EoS are indicated in Table AII.

For the determination of the incipient and complete melting entropy

TABLE AI
Mole Percentage of Oxide Components for Westerly Granite (USGS Sample G-1) and Mass Fraction of the Elements[a]

Oxide	Mole%	Element	Atomic no.	Mass fraction
SiO_2	72.64	O	8	0.62151
Al_2O_3	14.04	Si	14	0.2421
K_2O	5.48	Al	13	0.05616
Na_2O	3.32	K	19	0.03653
CaO	1.39	Na	11	0.02213
FeO	0.96	Ca	20	0.00695
Fe_2O_3	0.87	Fe	26	0.00828
H_2O	0.4	(H)	2	0.002667
MgO	0.38	(Mg)	12	0.0019
TiO_2	0.26	(Ti)	22	0.000867

[a] The elements not included in ANEOS are in parentheses.

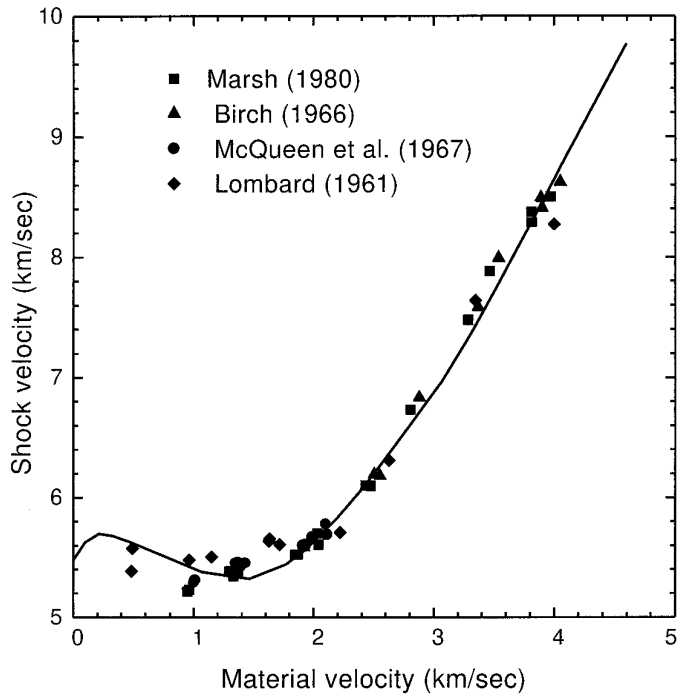

FIG. A1. Comparison of the Hugoniot for Westerly Granite calculated by ANEOS (solid line) with experimental data. A solid–solid phase transition breaks the shock–particle velocity linear relation in two separate linear segments, one for each solid phase.

TABLE AII
ANEOS Equation-of-State Parameters for Westerly Granite

Variable	Value	Description
Nel	7	Number of elements in the material
EoS	4	EoS type: solid–gas with electronic terms and detailed treatment of liquid/vapor region
ρ_0	2.63	Reference density (Westerly granite) (g/cm^3)
T_0	298	Reference temperature (K)
p_0	0	Reference pressure (dyn/cm^2)
S_0	5.5×10^5	Intercept of Hugoniot linear shock–particle velocity relation (cm/sec)
Γ_0	1.9	Reference Gruneisen coefficient
θ_0	0	Reference Debye temperature; set by ANEOS (0.025 eV)
S_1	1.56	Slope of linear Hugoniot shock–particle velocity relation
$3\gamma_0$	0	$3 \times$ limiting value of Gruneisen coefficient for large compressions
E_{sep}	2.48×10^{11}	Zero temperature separation energy (erg/g)
T_m	1673	Melting temperature (Allen 1967) (K)
c_{53}	10^{12}	Parameter for low-density modification of the critical point
c_{54}	0.7	Same as above
H_0	0	Parameter for thermal conduction (0 = not included)
c_{41}	0	Same as above
ρ_{min}	1	Lowest allowed solid density (g/cm^3)
D_1	4.7	ρ_1, density at beginning of the solid–solid phase transition (g/cm^3)
D_2	4.7	ρ_2, density at end of the solid–solid phase transition (g/cm^3)
D_3	4×10^{11}	Pressure at center of the phase transition at zero T (dyn/cm^2)
D_4	5×10^{11}	$dP/d\eta$ at high-pressure phase transition
D_5	5×10^{13}	$D^2P/d\eta^2$ at high-pressure phase transition
H_{fus}	0	Heat of fusion for melt transformation (not included)
ρ_{liq}	0	Ratio of liquid to solid density at melt point
U_p	0	Upper limit for cold compression relation for expanded states (def.)
L_0	0	Lower limit for cold compression relation for expanded states
α	0.5	Parameter for liquid EoS correction to match P_{vap} data and boiling point
β	0.35	Same as above
γ	0.5	Same as above
c_{60}	0.9	Interpolation parameter in Gruneisen coefficient model
c_{61}	0.3	Same as above
c_{62}	0.65	Interpolation parameter in free energy expression
Flag	1	Flag for ionization model: 0 = Saha, 1 = Thomas–Fermi
E_{sh}	0	Energy shift for reactive chemistry modeling
S_{sh}	0	Entropy shift for reactive chemistry modeling

of granite we used a thermodynamic model developed by Ghiorso and Sack (1995); the model is in the form of a FORTRAN code, called MELTS. Given the initial composition of the Westerly granite, the model provided the thermodynamic parameters associated with the melting curve at 1 bar. Since we assume an adiabatic release from the shock state to the reference state (1 bar), the entropies of incipient and complete melting can be used to determine the corresponding shock pressures on the Hugoniot.

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